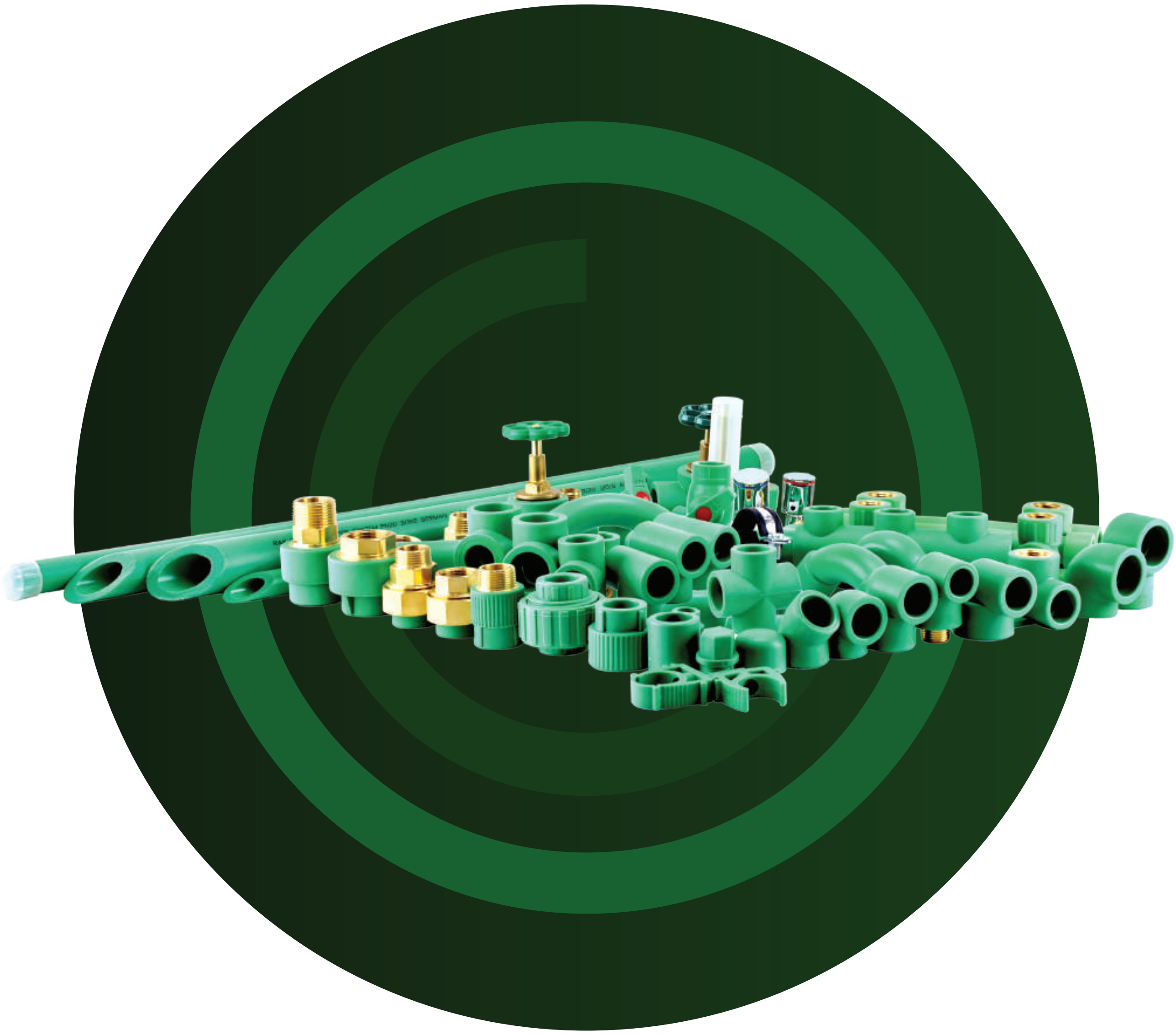
PP-R & PEX SYSTEMS
Reliable Piping System for Hot & Cold Water Transmission



EVOLVE
EXPAND
EMPOWER
through engineering


AQUASMART PLASTIC INDUSTRIES L.L.C
مصنع أكوا سمارت للبلاستيك ذ.م.م
EVOLVE EXPAND EMPOWER through engineering
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ABOUT US

Aquasmart Plastic Industries L.L.C is an ISO 9001-2015 certified company and now introducing its brand **RAKPLUS®** for PPR pipes and fittings, with dedication, trust and enthusiast.

Aquasmart Plastic Industries L.L.C has achieved a remarkable success in the region of GCC by its longest presence and commitment to the quality and service to the utmost backup of highly experienced and professional personal, which has enabled us to contribute significantly in the regional development of irrigation, construction, plumbing and landscape sectors.

Our Products are manufactured by using modern technology in the field of pipe extrusion and fittings in accordance with German standard.

The manufacturing process especially take care with in-house quality control to ensure that the product meets quality, reliability and accuracy for the better product and service to our valuable customers. Besides in-house quality control, tests are also carried out through independent laboratories of international standards to certify the quality of products.

We Strive to ensure that our customers receive the material in a timely and cost effective manner.

MISSION STATEMENT

We at **Aquasmart Plastic Industries L.L.C** think ourselves as a different kind of company, doing our part to make the world a better place. This pursuit is realized through unwavering commitment to leadership in everything we do. We don't just see to the needs of our customers; we also strive endlessly to surpass them. We don't just produce excellent products; we also work for continuous improvement. We don't just talk about quality; we also endorse quality management system to maintain ISO Certification in our organization and ensure our product meets or exceeds the most stringent quality standards.

Moreover, we do this all with respect for the beauty and fragility of the world and the well-being of its inhabitants, ensuring that our business makes a positive contribution to the environment, the communities we serve, our customers and our employees.

TECHNICAL INFORMATION

RAKPLUS® is the brand which made by modern innovation, latest German technology and standards. In accordance with its areas of application, the **RAKPLUS®** pipe system is designed for continuous temperatures of 0 °C to 70 °C, short-term peak temperatures of up to 100 °C and a service life of min. 50 years.

APPLICATIONS

- Hot & Cold potable water supply in residential and commercial buildings
- HVAC and compressed air system
- Pipe networks for rainwater utilization system
- Pipe networks for agricultural & horticultural use
- Pipe networks for swimming pool facilities
- Pipe networks for aggressive fluids (acidic, alkaline & corrosive chemicals etc.)

QUALITY

RAKPLUS® are designed to meet the harsh climate conditions of the GCC region and places emphasis on Quality, Reliability and Economy.

RAKPLUS® follow strict in-house Quality Control and backed by testing through independent laboratories of international repute to certify the quality of pipes and fittings.

RAKPLUS® do great emphasis on customer satisfaction through quality products. The company's operational excellence is evident through its established Quality management system which complies with ISO 9001 - 2015 standard, certified by quality Accreditation Bureau for Qualified Companies (QA QC), USA.

STANDARDS

Standards Applied in Production:
DIN 8077 / 8078 PP Pipes general quality requirements and testing.
DIN 8076 Pressure pipes consisting of Polyethylene metal clamping joint.
DVS 2207 Welding regulations for plastic pipes.
EN ISO 15874-2

THERMAL CONDUCTIVITY

The thermal conductivity of the pipes and fittings shall not exceed 0,24 W/mK at 20°C for water.

RAW MATERIALS

The basic material of the **RAKPLUS®** pipe system is polypropylene random-copolymer PP-R. This material is characterized by excellent properties such as elasticity, rigidity, tightness, compression strength and a special resistance to high temperatures and extraction. In addition, PP-R possesses high resistance to a large number of materials which might pass through the pipe system in liquid or gaseous form. This material is particularly suitable for use in drinking water systems

MECHANICAL AND THERMAL PROPERTIES

Properties	Standard	Typical value	Unit
Density	ISO 1183	905	Kg/m3
Melt Flow Rate	ISO 1133	≤ 0.3	g/10min
Flexural Modulus (2mm/min)	ISO 178	850	MPa
Tensile Modulus (1mm/min)	ISO 527	800	MPa
Tensile Strain at Yield (50mm/min)	ISO 527-2	13.5	%
Tensile Stress at Yield (50mm/min)	ISO 527-2	25	MPa
Thermal Conductivity	DIN 52612	0.24	W/(m K)
Coefficient of Thermal Expansion (0°C/70°C)	DIN 53752	1.8*10-4 1/K	
Charpy Impact Strength, notched (23° C)	ISO 179/1eA	60	
Charpy Impact Strength, notched (0° C)	ISO 179/1eA	6.0	
Charpy Impact Strength, Unnotched (23° C)	ISO 179/1eU	No Break	
Charpy Impact Strength, Unnotched (0° C)	ISO 179/1eU	No Break	

ADVANTAGES

- Very long life time with a guaranteed service life of 50 years
- Cost effective pipeline networks
- High impact strength & flexibility
- Damage resistance
- Damped vibrations and sounds are absorbed resulting to noise reduction
- Light weight, easy to install and low labour cost of installations
- Resistant to corrosion as compared to metal products
- Environment-friendly as it takes less energy to manufacture PPR pipes and are recyclable compared to its metal counterparts
- Hygienic and non-toxic compared to metal or other plastic products

PRODUCT RANGE AND COLOUR

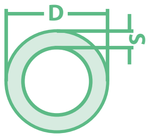
The **RAKPLUS®** pipe system is constantly being extended and updated to meet the requirements of the industry. Please refer to current RAKPLUS® part list for complete product range.

The colour of **RAKPLUS®** Products are available according to the standards and custom colour

PRODUCT SPECIFICATION

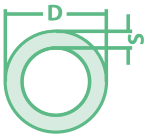
PIPE

Parameters of PPR Pipe SDR11/PN10 in accordance to DIN 8077/78



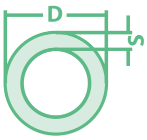
PART	DIMENSIONS	WALL THICKNESS	INNER DIAMETER	PACKING UNIT	Kg/Mtr.
RP1020	20	1.9	16.0	100Mtrs	0.107
RP1025	25	2.3	20.4	100Mtrs	0.164
RP1032	32	2.9	26.2	40Mtrs	0.261
RP1040	40	3.7	32.6	40Mtrs	0.412
RP1050	50	4.6	40.8	20Mtrs	0.638
RP1063	63	5.8	51.4	20Mtrs	1.010
RP1075	75	6.8	61.4	16Mtrs	1.410
RP1090	90	8.2	73.6	12Mtrs	2.030
RP10110	110	10.0	90.0	8Mtrs	3.010

Parameters of PPR Pipe SDR7.4/PN16 in accordance to DIN 8077/78



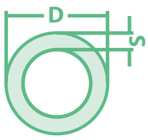
PART	DIMENSIONS	WALL THICKNESS	INNER DIAMETER	PACKING UNIT	Kg/Mtr.
RP1620	20	2.8	14.4	100Mtrs	0.147
RP1625	25	3.5	18.0	100Mtrs	0.229
RP1632	32	4.4	23.2	40Mtrs	0.370
RP1640	40	5.5	29.0	40Mtrs	0.578
RP1650	50	6.9	36.2	20Mtrs	0.906
RP1663	63	8.6	45.8	20Mtrs	1.426
RP1675	75	10.3	54.4	16Mtrs	2.031
RP1690	90	12.3	65.4	12Mtrs	2.880
RP16110	110	15.1	79.8	8Mtrs	4.295

Parameters of PPR Pipe SDR6/PN20 in accordance to DIN 8077/78



PART	DIMENSIONS	WALL THICKNESS	INNER DIAMETER	PACKING UNIT	Kg/Mtr.
RP2020	20	3.4	13.2	100Mtrs	0.172
RP2025	25	4.2	16.6	100Mtrs	0.266
RP2032	32	5.4	21.2	40Mtrs	0.432
RP2040	40	6.7	26.6	40Mtrs	0.670
RP2050	50	8.3	33.4	20Mtrs	1.040
RP2063	63	10.5	42.0	20Mtrs	1.645
RP2075	75	12.5	50.0	16Mtrs	2.340
RP2090	90	15.0	60.0	12Mtrs	3.360
RP20110	110	18.3	73.4	8Mtrs	5.008

Parameters of PPR Pipe SDR5/PN25 Series 2 in accordance to DIN 8077/78



PART	DIMENSIONS	WALL THICKNESS	INNER DIAMETER	WATER CONTENT (l/mtr)	PACKING UNIT	Kg/Mtr.
RP2520	20	4.1	11.8	0.111	100Mtrs	0.198
RP2525	25	5.1	14.8	0.178	100Mtrs	0.307
RP2532	32	6.5	19.0	0.291	40Mtrs	0.498
RP2540	40	8.1	23.8	0.461	40Mtrs	0.775
RP2550	50	10.1	29.8	0.703	20Mtrs	1.210
RP2563	63	12.7	37.6	1.137	20Mtrs	1.910
RP2575	75	15.1	44.8	1.619	16Mtrs	2.701
RP2590	90	18.1	53.8	2.336	12Mtrs	3.880
RP25110	110	22.1	65.8	2.742	8Mtrs	5.780

PRODUCT SPECIFICATION

FITTINGS

SDR6/PN20

SOCKET

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPS - 20	20mm	20	200
RPS - 25	25mm	20	200
RPS - 32	32mm	10	200
RPS - 40	40mm	5	150
RPS - 50	50mm	5	100
RPS - 63	63mm	1	50
RPS - 75	75mm	1	50
RPS - 90	90mm	1	40
RPS - 110	110mm	1	30

REDUCER SOCKET

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPRS - 2520	25/20mm	25	200
RPRS - 3220	32/20mm	25	200
RPRS - 3225	32/25mm	20	200
RPRS - 4020	40/20mm	5	200
RPRS - 4025	40/25mm	5	150
RPRS - 4032	40/32mm	5	150
RPRS - 5020	50/20mm	5	150
RPRS - 5025	50/25mm	5	100
RPRS - 5032	50/32mm	5	100
RPRS - 5040	50/40mm	5	100
RPRS - 6320	63/20mm	1	100
RPRS - 6325	63/25mm	1	100
RPRS - 6332	63/32mm	1	75
RPRS - 6340	63/40mm	1	75
RPRS - 6350	63/50mm	1	75
RPRS - 7540	75/40mm	1	50
RPRS - 7550	75/50mm	1	50
RPRS - 7563	75/63mm	1	50
RPRS - 9050	90/50mm	1	50
RPRS - 9063	90/63mm	1	50
RPRS - 9075	90/75mm	1	40
RPRS - 11063	110/63mm	1	40
RPRS - 11075	110/75mm	1	40
RPRS - 11090	110/90mm	1	40

SDR6/PN20

Elbow 90°

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPE90 - 20	20mm	10	200
RPE90 - 25	25mm	10	400
RPE90 - 32	32mm	5	200
RPE90 - 40	40mm	5	150
RPE90 - 50	50mm	5	100
RPE90 - 63	63mm	1	50
RPE90 - 75	75mm	1	30
RPE90 - 90	90mm	1	20
RPE90 - 110	110mm	1	10

Elbow 45°

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPE45 - 20	20mm	10	200
RPE45 - 25	25mm	10	400
RPE45 - 32	32mm	5	200
RPE45 - 40	40mm	5	150
RPE45 - 50	50mm	5	100
RPE45 - 63	63mm	1	50
RPE45 - 75	75mm	1	30
RPE45 - 90	90mm	1	20
RPE45 - 110	110mm	1	10

Reducer Elbow

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPRE - 2520	20mm	10	200
RPRE - 3225	25mm	10	250
RPRE - 4032	32mm	5	150

Spigot Elbow 90°

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPSE - 3225	25mm	10	300
RPSE - 4032	32mm	5	200

SDR6/PN20

Reducer Tee

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPRT - 202520	20X25X20mm	10	250
RPRT - 252020	25X20X20mm	10	200
RPRT - 252025	25X20X25mm	10	150
RPRT - 253225	25X32X25mm	10	150
RPRT - 322032	32X20X32mm	5	100
RPRT - 322525	32X25X25mm	5	100
RPRT - 323225	32X32X25mm	5	100
RPRT - 402040	40X20X40mm	5	75
RPRT - 402540	40X25X40mm	5	75
RPRT - 403240	40X32X40mm	5	75
RPRT - 502550	50X25X50mm	5	50
RPRT - 502550	50X25X50mm	5	50
RPRT - 503250	50X32X50mm	5	50
RPRT - 504050	50X40X50mm	5	50
RPRT - 632063	63X20X63mm	1	40
RPRT - 632563	63X25X63mm	1	40
RPRT - 633263	63X32X63mm	1	40
RPRT - 634063	63X40X63mm	1	40
RPRT - 635063	63X50X63mm	1	30
RPRT - 753275	75X32X75mm	1	25
RPRT - 754075	75X40X75mm	1	25
RPRT - 755075	75X50X75mm	1	25
RPRT - 756375	75X63X75mm	1	20
RPRT - 904090	90X40X90mm	1	15
RPRT - 905090	90X50X90mm	1	15
RPRT - 906390	90X63X90mm	1	15
RPRT - 907590	90X75X90mm	1	15
RPRT - 11050110	110X50X110mm	1	10
RPRT - 11063110	110X63X110mm	1	10
RPRT - 11075110	110X75X110mm	1	10
RPRT - 11090110	110X90X110mm	1	10

Cross Tee

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPXT - 20	20mm	10	150
RPXT - 25	25mm	10	200
RPXT - 30	30mm	5	150

SDR6/PN20

Equal Tee

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPT - 20	20mm	10	200
RPT - 25	25mm	10	150
RPT - 32	32mm	5	150
RPT - 40	40mm	5	100
RPT - 50	50mm	5	75
RPT - 63	63mm	1	50
RPT - 75	75mm	1	25
RPT - 90	90mm	1	15
RPT - 110	110mm	1	10

Union Both Ends Welding

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPU - 20	20mm	10	100
RPU - 25	25mm	10	100
RPU - 32	32mm	5	50

Cross Over Long Type

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPCOL - 20	20mm	10	150
RPCOL - 25	25mm	5	150
RPCOL - 32	32mm	5	75

Cross Over Short Type

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPU - 20	20mm	10	300
RPU - 25	25mm	10	300

SDR6/PN20

End Cap

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPEC - 20	20mm	25	400
RPEC - 25	25mm	25	250
RPEC - 32	32mm	20	200
RPEC - 40	40mm	10	150
RPEC - 50	50mm	10	100
RPEC - 63	63mm	5	100
RPEC - 75	75mm	5	75
RPEC - 90	90mm	1	50
RPEC - 110	110mm	1	25

Saddle Socket

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPSS - 5025	50X25mm	10	200
RPSS - 5032	50X32mm	10	200
RPSS - 6320	63X20mm	5	150
RPSS - 6325	63X25mm	5	150
RPSS - 6332	63X32mm	5	150
RPSS - 7520	75X20mm	2	100
RPSS - 7525	75X25mm	2	100
RPSS - 7532	75X32mm	2	100
RPSS - 9020	90X20mm	1	75
RPSS - 9025	90X25mm	1	75
RPSS - 9032	90X32mm	1	75
RPSS - 11020	110X20mm	1	50
RPSS - 11025	110X25mm	1	50
RPSS - 11032	110X32mm	1	50

Bend

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPB - 20	20mm	10	150
RPB - 25	25mm	10	200
RPB - 32	32mm	5	100

SDR6/PN20

END PLUG TYPE A (SHORT)

PART NO	DIMENSION	PACKING UNIT	
		Pcs/Packet	Pcs/Box
RPEPA - ½"	½"	100	1500
RPEPA - ¾"	¾"	100	1000
RPEPA - 1"	1"	50	500

END PLUG TYPE B (LONG)

PART NO	DIMENSION	PACKING UNIT	
		Pcs/Packet	Pcs/Box
RPEPB - ½"	20mm	50	400
RPEPB - ¾"	25mm	50	300

BRACKET FOR PIPE

PART NO	DIMENSION	PACKING UNIT	
		Pcs/Packet	Pcs/Box
RPBT - 20	20mm	100	1500
RPBT - 25	25mm	100	1000
RPBT - 32	32mm	100	500
RPBT - 40	40mm	50	300

BRACKET WITH CLAMP

PART NO	DIMENSION	PACKING UNIT	
		Pcs/Packet	Pcs/Box
RPBT - 25	25mm	100	500
RPBT - 32	32mm	50	500

SDR6/PN20

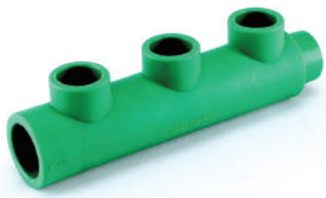
Pipe Hanging Clamp

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPHC - 20	20mm	100	200
RPHC - 25	25mm	100	200
RPHC - 32	32mm	50	100
RPHC - 40	40mm	50	100

PPR Ball Valve - Lever Type

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPBVL - 20	20mm	5	75
RPBVL - 25	25mm	5	75
RPBVL - 32	32mm	5	50
RPBVL - 40	40mm	2	30
RPBVL - 50	50mm	2	20
RPBVL - 63	63mm	2	10

PPR Manifold



Part No	Dimension	Packing Unit
		Pcs/Packet Pcs/Box
RPMF 3225 - 2W	32 X 25 mm (Two Way)	100
RPMF 3225 - 3W	32 X 25 mm (Three Way)	100
RPMF 3225 - 4W	32 X 25 mm (Four Way)	100
RPMF 4032 - 2W	40 X 32 mm (Two Way)	75
RPMF 4032 - 3W	40 X 32 mm (Three Way)	75
RPMF 4032 - 4W	40 X 32 mm (Four Way)	75
RPMF 5025 - 4W	50 X 25 mm (Four Way)	30
RPMF 5032 - 4W	50 X 32 mm (Four Way)	30
RPMF 6322 - 4W	63 X 25 mm (Four Way)	10
RPMF 6332 - 4W	63 X 32 mm (Four Way)	10

Transition Female Socket-Round

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPFS - 20½”	20mm X ½”	10	200
RPFS - 20¾”	20mm X ¾”	10	200
RPFS - 25½”	25mm X ½”	10	200
RPFS - 25¾”	25mm X ¾”	10	200
RPFS - 32¾”	32mm X ¾”	5	100

SDR6/PN20

TRANSITION FEMALE SOCKET-HEXAGONAL

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPFSH - 321	32mm X 1"	5	50
RPFSH - 401	40mm X 1"	5	50
RPFSH - 401¼	40mm X 1¼"	5	50
RPFSH - 501½	50mm X 1½"	2	30
RPFSH - 632	63mm X 2"	2	25
RPFSH - 752½	75mm X 2½"	1	20
RPFSH - 903	90mm X 3"	1	10
RPFSH - 1104	110mm X 4"	1	5

TRANSITION MALE SOCKET-ROUND

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPMS - 20½	20mm X ½"	10	200
RPMS - 20¾	20mm X ¾"	10	200
RPMS - 25½	25mm X ½"	10	200
RPMS - 25¾	25mm X ¾"	10	200
RPMS - 32¾	32mm X ¾"	5	150
RPMS - 321	32mm X 1"	5	50
RPMS - 501½	50mm X 1½"	2	50

TRANSITION MALE SOCKET-HEXAGONAL

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPMSH - 321	32 X 1"	5	50
RPMSH - 401	40 X 1"	5	50
RPMSH - 401¼	40 X 1¼"	5	40
RPMSH - 501½	50 X 1½"	2	40
RPMSH - 632	63 X 2"	2	25
RPMSH - 752½	75 X 2½"	1	15
RPMSH - 903	90 X 3"	1	10
RPMSH - 1104	110mm X 4"	1	5

TRANSITION FEMALE UNION-BRASS

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPFU - 201½	20mm X 1½"	10	100
RPFU - 25¾	25mm X ¾"	10	100
RPFU - 321	32mm X 1"	5	50
RPFU - 401¼	40mm X 1¼"	2	40
RPFU - 501½	50mm X 1½"	2	30
RPFU - 632	63mm X 2"	1	15

SDR6/PN20

Transition Male Union-Brass

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPMU - 20½	20mm X ½"	10	100
RPMU - 25¾	25mm X ¾"	10	100
RPMU - 321	32mm X 1"	5	50
RPMU - 401¼	40mm X 1¼"	2	40
RPMU - 501½"	50mm X 1½"	2	30
RPMU - 632	63mm X 2"	1	15

Transition Female Tee - Round/Hexagon

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPFT - 20½	20mm X ½"	10	100
RPFT - 20¾	20mm X ¾"	10	100
RPFT - 25½	25mm X ½"	10	150
RPFT - 25¾	25mm X ¾"	10	150
RPFT - 32½	32mm X ½"	5	100
RPFT - 32¾	32mm X ¾"	5	100
RPFT - 321	32mm X 1"	5	50
RPFTH - 321	32mm X 1"	5	50

Transition Valve Body

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPVB - 20¾	20mm X ¾"	10	100
RPVB - 25¾	25mm X ¾"	10	100
RPVB - 321	32mm X 1"	5	50
RPVB - 401¼	40mm X 1¼"	2	50

Transition Female Elbow - Round/Hexagon

Part No	Dimension	Packing Unit Pcs/Packet	Packing Unit Pcs/Box
RPFE - 20½	20mm X ½"	10	100
RPFE - 20¾	20mm X ¾"	10	100
RPFE - 25½	25mm X ½"	10	200
RPFE - 25¾	25mm X ¾"	10	200
RPFE - 32½	32mm X ½"	5	100
RPFE - 32¾	32mm X ¾"	5	75
RPFE - 321	32mm X 1"	5	50
RPFEH - 321	32mm X 1"	5	50

SDR6/PN20

Transition Male Elbow

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPFE - 20½	20mm X ½"	10	100
RPFE - 20¾	20mm X ¾"	10	100
RPFE - 25½	25mm X ½"	10	200
RPFE - 25¾	25mm X ¾"	10	200
RPFE - 32½	32mm X ½"	5	100
RPFE - 32¾	32mm X ¾"	5	75
RPFE - 321	32mm X 1"	5	50

Transition Female Wall Elbow

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPFWE - 20½	20mm - X ½"	10	100
RPFWE - 25½	25mm - X ½"	10	100

Transition Wall Mount Group-Horizontal

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPWMGH - 20	20 mm	5	30
RPWMGH - 25	25 mm	5	40

Transition Wall Mount Group-Vertical

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPWMGV - 25	25 mm	5	40

SDR6/PN20

Stop Valve With Round Wheel

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPSV - 20	20mm	1	50
RPSV - 25	25mm	1	50
RPSV - 32	32mm	1	50
RPSV - 40	40mm	1	50

Concealed Valve

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPCV - 20	20mm	1	50
RPCV - 25	25mm	1	50
RPCV - 32	32mm	1	40

Ball Valve - Brass

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPBBV - 20	20mm	1	30
RPBBV - 25	25mm	1	30
RPBBV - 32	32mm	1	15
RPBBV - 40	40mm	1	15
RPBBV - 50	50mm	1	10
RPBBV - 63	63mm	1	10

Angle Valve - V5

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPAV5 - ½	½" X ½"	1	100

Angle Valve - V8

Part No	Dimension	Packing Unit	
		Pcs/Packet	Pcs/Box
RPAV8 -½"	½" X ½"	1	100

MAXIMUM OPERATING PRESSURE (MOP) FOR PP-R, FOR WATER SAFETY FAOTOR (SF) = 1,25 DIN 8077

Table (01)

Temperature °C	Operating Years	SDR 11	SDR 7,4	SDR 6	SDR 5
		Maximum Operating Pressure			
10	1	21,01	33,4	42,1	53,0
	5	19,81	31,5	39,7	49,9
	10	9,03	30,7	38,6	48,7
	25	18,07	29,7	37,4	47,0
	50	18,02	28,9	36,4	45,9
	100	17,08	28,2	35,5	44,7
20	1	18,00	28,5	35,9	45,2
	5	16,09	26,8	33,7	42,5
	10	16,04	26,1	32,8	41,4
	25	15,09	25,2	31,7	39,9
	50	15,04	24,5	30,9	38,9
	100	15,00	23,9	30,2	37,8
30	1	15,03	24,2	30,5	38,5
	5	14,03	22,7	28,6	36,0
	10	13,09	22,1	27,8	35,0
	25	13,04	21,3	26,8	33,8
	50	13,00	20,7	26,1	32,9
	100	12,07	20,1	25,4	31,9
40	1	13,00	20,6	25,9	32,6
	5	12,01	19,2	24,2	30,5
	10	11,08	18,7	23,5	29,6
	25	11,03	18,0	22,6	28,5
	50	11,00	17,4	22,0	27,7
	100	10,07	16,9	21,4	26,9
50	1	11,00	17,4	21,9	27,6
	5	10,02	16,2	20,4	25,7
	10	9,09	15,7	19,8	25,0
	25	9,05	15,1	19,0	24,0
	50	9,02	14,7	18,5	23,3
	100	9,00	14,2	17,9	22,6
60	1	9,02	14,7	18,5	23,3
	5	8,06	13,6	17,2	21,6
	10	8,03	13,2	16,6	21,0
	25	8,00	12,7	16,0	20,1
	50	7,07	12,3	15,5	19,5
70	1	7,08	12,3	15,5	19,6
	6	7,02	11,4	14,4	18,1
	10	7,00	11,1	13,9	17,5
	25	6,00	9,6	12,1	15,2
	60	5,01	8,1	10,2	12,8
80	1	6,05	10,3	13,0	16,4
	5	5,07	9,1	11,5	14,5
	10	4,08	7,7	9,7	12,2
	25	3,09	6,2	7,8	9,8
95	1	4,06	7,3	9,2	11,6
	6	3,01	4,9	6,2	7,8
	(10)	(2,06)	(4,1)	(5,2)	(6,6)

Data inside the brackets apply by verification at longer testing periods than 1 year at the 110 °C-test.

Maximum Operating Pressure

Table (02)

Heating period	Temperature °C	Years	SDR 9	SDR 7,4	SDR 5
			Maximum Operating Pressure		
Continuous operating temperature 70 °C incl. 60 days per year at...	75°C	5	14,30	11,40	15,90
		10	13,70	11,90	14,50
		25	11,80	9,30	13,70
		45	10,40	8,10	12,80
	80°C	5	12,90	10,07	15,80
		10	12,20	9,70	15,40
		25	10,70	8,60	13,20
		40	9,80	7,80	11,60
	85°C	5	12,51	9,94	15,78
		10	11,90	9,50	15,30
		25	9,70	7,80	13,20
		35	8,90	7,10	11,20
	90°C	5	11,80	9,37	14,90
		10	10,30	8,40	12,48
		25	8,40	6,60	10,48
		30	7,63	6,30	8,45
Continuous operating temperature 70 °C incl. 90 days per year at...	75°C	5	13,95	11,50	14,73
		10	13,40	10,80	13,80
		25	11,50	9,20	12,40
		45	8,90	7,00	11,20
	80°C	5	12,75	10,14	16,10
		10	12,33	9,81	15,50
		25	10,06	8,02	12,71
		37,5	9,15	7,27	11,52
	85°C	5	12,00	9,54	15,15
		10	11,29	9,00	14,20
		25	9,62	7,63	12,16
		32,5	9,07	7,20	11,40
	90°C	5	10,79	8,60	11,30
		10	9,30	7,41	10,45
		25	7,35	5,73	9,22

* SDR = Standard Dimension Ratio (=Diameter/Wall thickness)

For having enough elasticity, in application the leught of bending side of the pipe is important. This can be calculated as fallows.

Ls = K.√d.ΔL in this formula
Ls = Leught of free bending piece L=Leught of pipe
K = Constant coefficient for pipes = 30
ΔL = Elongation or shringkage in mm
d = Outside diameter of RAK PLUS pipe in (mm)

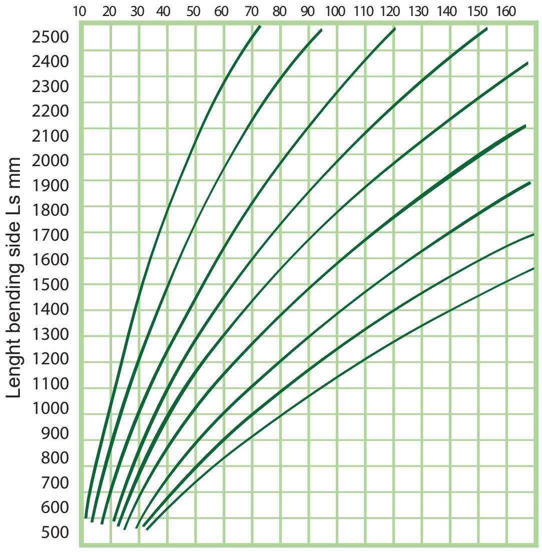
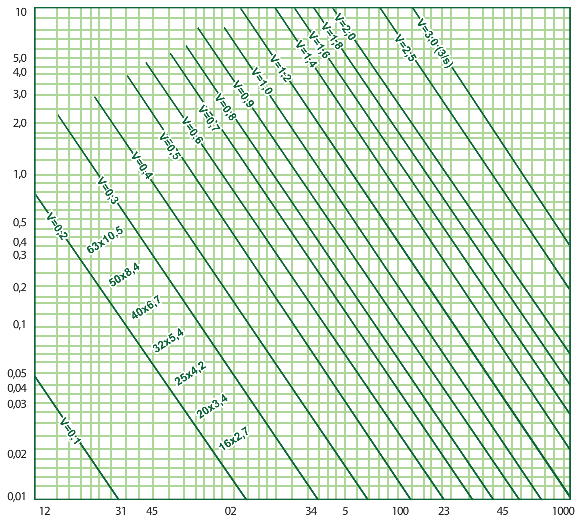


Figure 8: Lenght bending side (Ls) depending on extention

PRESSURE DROPS



Calculation of thermal Expansion RAKPLUS®

Expansion of the polyprolyene pipes are relatively higher and linear expansion coeftficient is 10 times bigger than metal pipes. That's why in installation this expansion character must be taken into consideration.Linear enpansion coefficient of **RAKPLUS®** pipes: temperature is between 30-90ºC the expansion ΔL is calculated with the folloving formula:

$\Delta = \lambda \times l \Delta\ T$ Δl= Linear expansion in mm

λ = Linear expansion coefficient = $\frac{mm}{m.^\circ C}$

RAKPLUS® pipe (Averere value) =0.183 $\frac{mm}{m.K}$

l= Pipe lenght in m
ΔT=Temperature difference Kº or Cº

ΔT=Difference of temperature between hot water and ambiant temperature in ˚C

Example:
Pipe temperature at the first installation is: + 16 ˚C and pipe lenght is 8 m
Minimum pipe temperature (for cold water) : + 9 ˚C
Maximum pipe temperature (for hot water) : + 70 ˚C

1 - Difference between pipe temperature at the first installation and minimum pipe temperature.
ΔT1= 9-16= -7 ºC

2 - Difference between pipe temperature at the first installation and maximum pipe temperature.
ΔT2= 70-16= 54 ºC

Expansion of pipe for ΔT1
ΔL1= 8m (-7ºC).0,183 mm/m= -10,2 mm
Expansion of pipe for T2
ΔL2= 8m * 54 ºC * 0,183 mm/m ºC= 79.0 mm

Application and Their Pressure Readings

Minimum Pressure of flow P min F1 BAR	TYPE OF THE POINT	Calgulation of flow during intake: TYPE OF THE POINT 1) Mixed water V R COLD	V R HOT	Only cold or heated potable water V R
	DESIGNATION Taps:	l/s	l/s	l/s
0.5	2) Without inlet DN	-	-	0,03
0.5	2) Without inlet DN	-	-	0,05
0.5	2) Without inlet DN	-	-	1
1	2) Without inlet DN	-	-	0,15
1	2) Without inlet DN	-	-	0,15
1	Shower heads for shower	0,01	0,01	0,02
1,2	Flush valvesacc. To DN 3265 DN	-	-	0,07
1,2	Part 1 DN	-	-	1
0,4	DN	-	-	1
1	Siphon for toilet DN	-	-	0,03
1	Dish washer DN	-	-	0,15
1	Washing Machine DN	-	-	0,25
	Battery:			
1	Shower DN 15	0,15	0,15	-
1	Bath tub DN 15	0,15	0,15	-
1	Kitchen Sink DN 15	0,07	0,07	-
1	Wash Stand DN 15	0,07	0,07	-
1	For small bath tub DN 15	0,07	0,07	-
1	Battery DN 20	0,03	0,03	-
0,5	DIN19542 kitchen sink DN 15	-	-	0,13
1	Electric water boiler DN 15	-	-	0,10

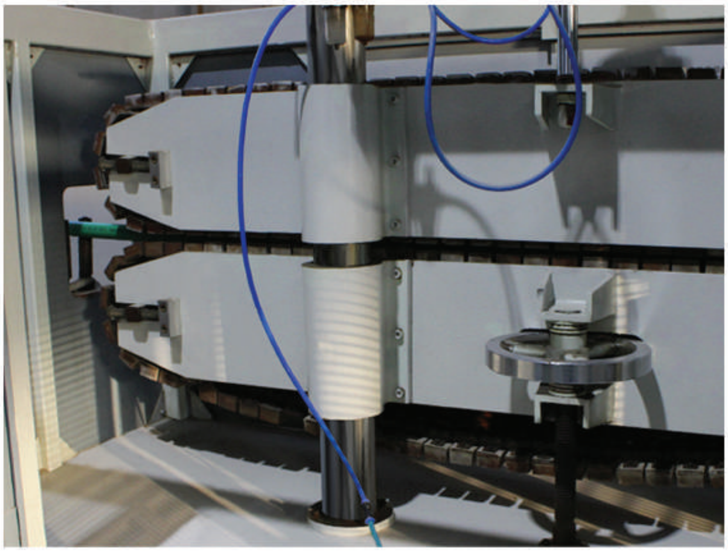
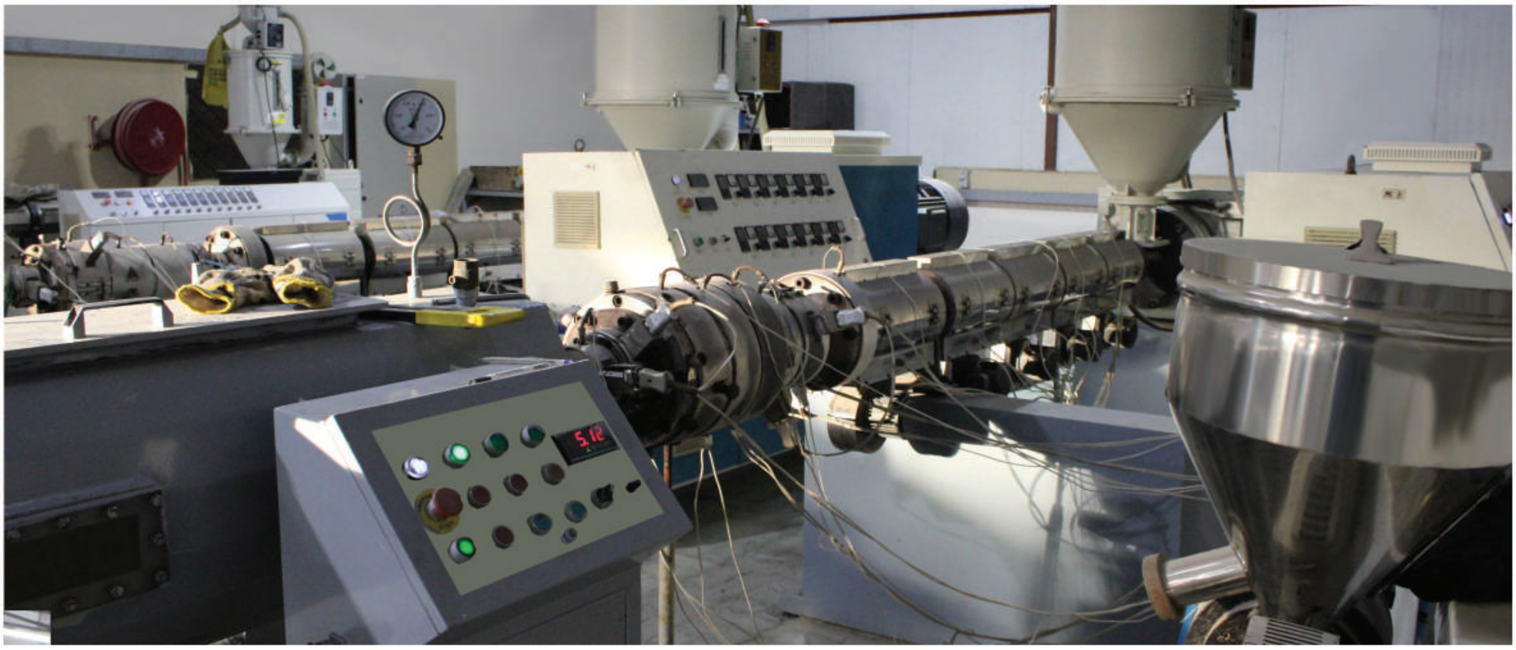
P.N. Resources and machine that are not listed in the above table must be considered, big armature flows or those given as minimum flow pressure must be calculated according to the data given by the manufacturer,

1) For mixed water resources, the flow is 15° C for cold water and 60° C for lukewarm drinking water.
2) Water without jet and valves with lo meter hose attachment, or in water tankers, the loss of pressure will be calculated according to minimum flow pressure on lump sum basis. In this case, the minimum flow pressure will be raised from 1.0 to 1.5 bars.
3) Fully opened water faucet.

Chemical Resistance Chart

Reagent	Concentration	Temperature 20°C	60°C	100°C
acetic anhydride	100%	G	-	-
acetic acid:di tri chloro acetic acid	sol	G	-	-
acetic acid	up to 40 %	G	G	-
acetic acid	50 %	G		G
acetic glacial acid	over 96 %	G	S	N
acetone	100 %	G	S	
acetophenone anhydride	100 %	G	S	-
acrylonitrile	100 %	G	-	-
air		G	G	G
almond oil		G	-	-
alum	sol	G	-	-
ammonia (gas)	100 %	G	-	-
ammonia (saturated in water)		G	G	-
ammonia liquor	up to 30 %	G	G	-
ammonium acetate	sat.sol	G	G	-
ammonium bicarbonate	sat.sol	G	G	-
ammonium chloride	sat.sol	G	G	-
ammonium fluoride	sol.	G	G	-
ammonium hydroxide	sol.	G	-	-
ammonium metaphosphate	sat.sol.	G	G	G
ammonium nitrate	sat.sol	G	G	G
ammonium phosphate	sat.sol	G	G	-
ammonium sulphate	sat.sol	G	G	G
ammonium acetate	100 %	G	-	-
amyl alcohol	100 %	S	G	G
aniline	100 %	S	-	-
anisole	100 %	G	-	-
apple juice		G	G	-
barium chloride	sat.sol.	G	G	G
barium carbonate	sat.sol.	G	G	G
barium hydroxide	sat.sol.	G	G	G
barium sulphate	sat.sol.	G	G	G
benzoic acid	sat.sol.	G	-	-
benzoyl acid		G	G	-
benzyl alcohol	100 %	G	S	-
borax	sol	G	G	-
boric acid	sat.sol.	G	-	
butane	100 %	G	G	-
butanol	100 %	G	S	S
butyl glycol	100 %	G	-	-
butylphenol cold	sat.sol.	G	-	-
butyl phthalate	100 %	G	S	S
calcium carbonate	sat.sol.	G	G	G
calcium chloride	sat.sol.	G	G	G
calcium hydroxide	sat.sol.	G	G	-
calcium nitrate	sat.sol.	G	G	-
carbon dioxide, gaseous, dry	100 %	G	G	-
carbon dioxide, gaseous, wat		G	G	-
carbon di-sulphide	100 %	NS	NS	NS

Reagent	Concentration	Temperature 20°C	60°C	100°C
carbon tetrachloride	100%	NS	NS	NS
castor-oil		100%	G	G
chloroethanol (2-Chloroethanol)	100%	G	-	-
chrome alum	sat.sol.	G	G	-
chromic acid	up to 40%	S	S	NS
citric acid	10%	G	G	G
coconut-oil	G	-	-	
corn-oil	G	S	-	
cotton-oil	G	S	-	
cresol	over 90%	G	-	-
cupric chloride	sat.sol.	G	G	-
cupric nitrate	30%	G	G	
cupric sulphate	sat.sol.	G	G	-
cyclohexane	100%	G	-	-
cyclohexanol	100%	G	S	-
dextrin	sol.	G	G	-
dextrose	sol.	G	G	-
dibutyl phthalate	100%	G	S	NS
dichloroethylene acid	100%	S	-	-
dichloroethylene	100%	S	-	-
diethanolamine	100%	G	-	-
diethyl ether	100%	G	S	-
diethylene glycol	100%	G	G	-
diglycolic acid	sat.sol.	G	-	-
diisooctyl phthalate	100%	G	S	-
dimethylamine	100%	G	-	-
dioctyl phthalate	100%	S	G	-
dioxane	100%	S	S	-
ethanolamine	100%	G	S	-
ethyl alcohol (ethanol)	up to 95%	G	-	-
ethylene chloride	100%	NS	G	-
ethyleenglycol	100%	G	NS	G
formaldehyde	40%	G	G	-
	10%	G	-	S
		G		

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